

Social Media Engagement Analytics

This project analyzed **5,000 social media records** using Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly to identify patterns in content performance, user behavior, device usage, country, and sentiment.

Key Findings

Content Performance: Video posts achieved the highest average engagement rate (**1.12**), making them the best-performing post type. Among content categories, **food** performed best with an average engagement rate of **1.36**, followed by technology (**1.16**).

Country Performance: **Brazil** recorded the highest average engagement rate (**1.54**), followed by Australia (**1.32**) and France (**1.15**). The USA had the lowest average engagement rate (**0.58**), showing considerable variation in engagement across countries.

User Trends: Verified accounts had a higher average engagement rate (**1.05**) than non-verified accounts (**0.95**). Verified accounts also recorded higher average impressions and watch time, although non-verified accounts had slightly higher average likes.

Device Behavior: **Mobile users** showed the highest average watch time (**4,087.83 seconds**) compared with tablet (**3,979.74 seconds**) and desktop (**3,974.79 seconds**). Mobile users also had the highest average engagement rate (**1.12**), indicating strong engagement through mobile devices.

Sentiment Analysis: **Negative sentiment posts** achieved the highest average engagement rate (**1.04**), followed by neutral (**0.99**) and positive (**0.92**) posts. Negative posts also had the highest average likes and shares, while neutral posts had the highest average comments.

Posting Time: The dataset could not be used to determine the best time of day for impressions because all 5,000 records had a posting hour of **0**. Therefore, there was insufficient variation in the timestamp data to draw a reliable conclusion about posting time.

Conclusion

Overall, the analysis shows that **video posts, food content, mobile usage, verified accounts, and negative-sentiment posts** were associated with stronger engagement measures. Brazil was the highest-performing country by average engagement rate. These findings provide useful insights for developing more effective, data-driven social media content strategies.